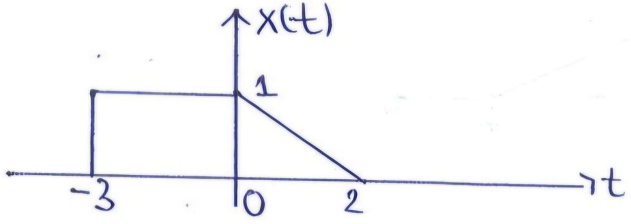
Shembulli 2: Të gjendet $x(-\frac{t}{2} - \frac{1}{2})$:

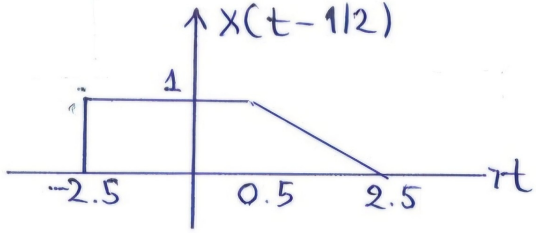
$x(-\frac{t}{2} - \frac{1}{2})$

1. sinjali $x(t)$
2. zhvendosja $x(t - \frac{1}{2})$
3. zgjerimi $x(\frac{t}{2} - \frac{1}{2})$
4. pasqyrimi $x(-\frac{t}{2} - \frac{1}{2})$

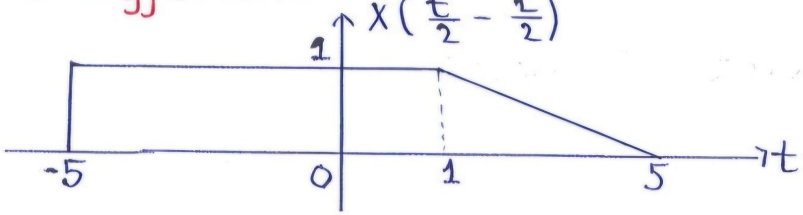
1. Sinjali:



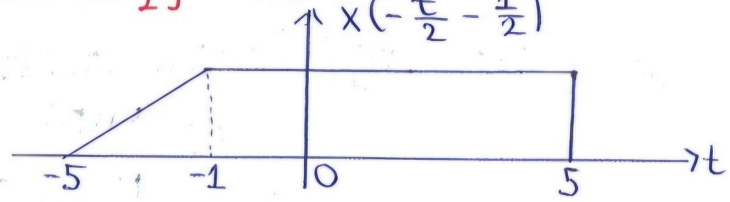
2. Zhvendosja:



3. Zgjerimi:

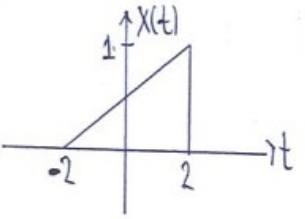


4. Pasqyrimi:

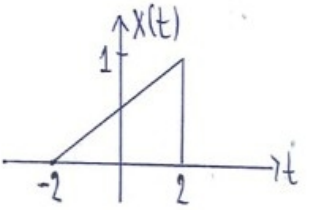


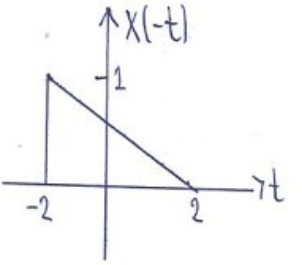
Detyra 1: Të gjenden komponentët çift dhe tek të sinjalit:

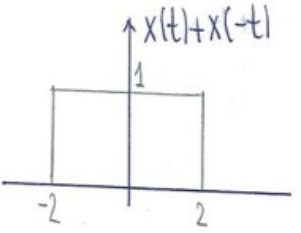
a)

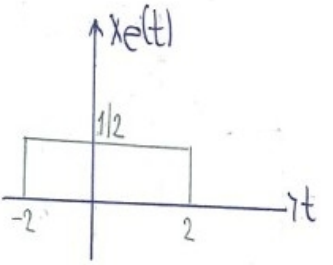


Pjesa çifte: $x_e(t) = \frac{x(t)+x(-t)}{2}$

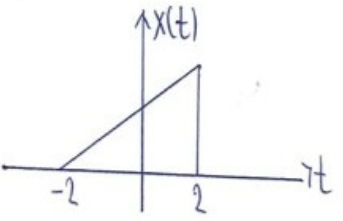


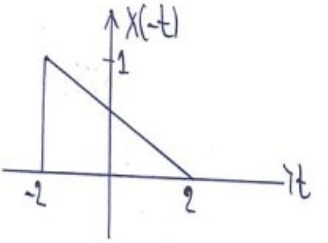


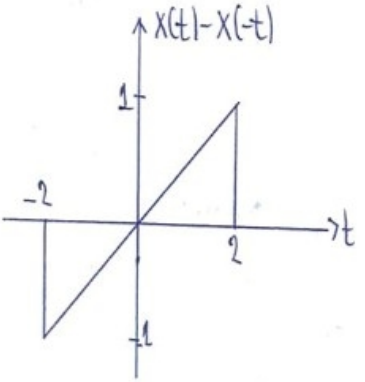


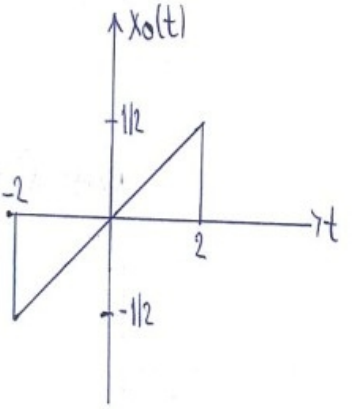


Pjesa teke: $x_o(t) = \frac{x(t)-x(-t)}{2}$

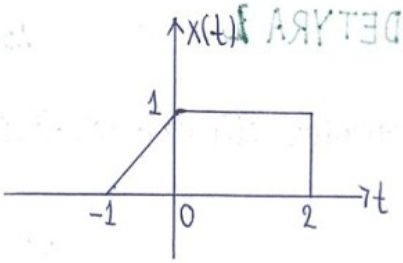




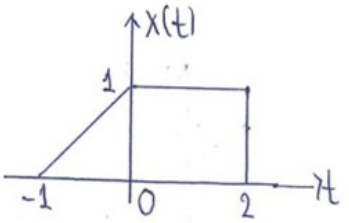


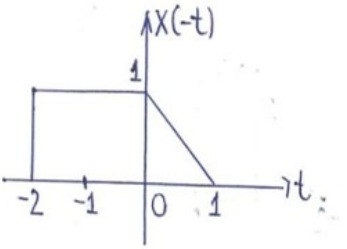


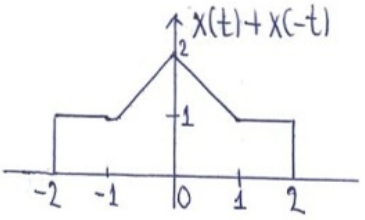
b)

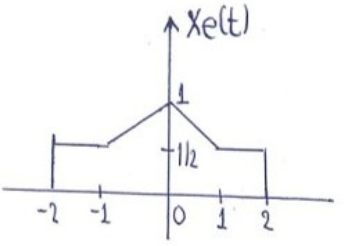


Pjesa çifte: $x_e(t) = \frac{x(t) + x(-t)}{2}$

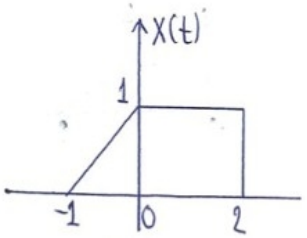


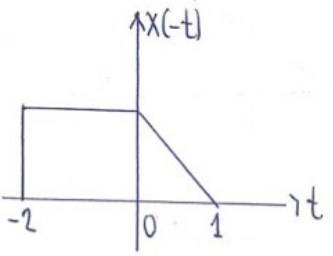


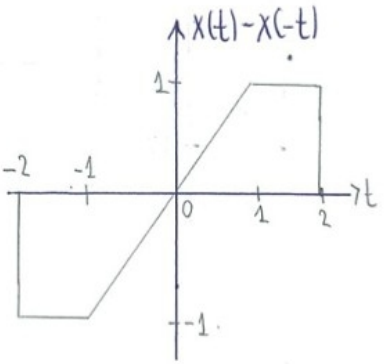


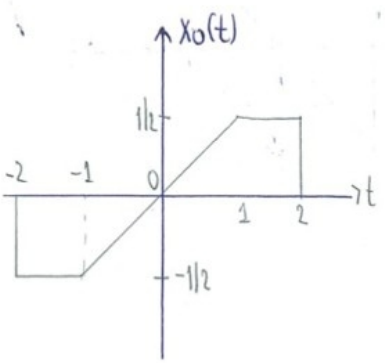


Pjesa teke: $x_o(t) = \frac{x(t) - x(-t)}{2}$

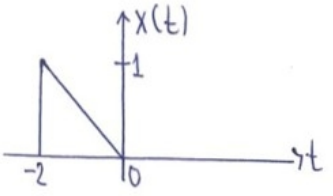




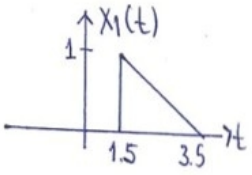




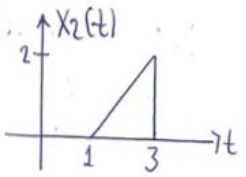
Detyra 4: Të shprehen sinjalet $x_1(t)$, $x_2(t)$, $x_3(t)$, $x_4(t)$, $x_5(t)$ dhe $x_6(t)$ në funksion të $x(t)$:



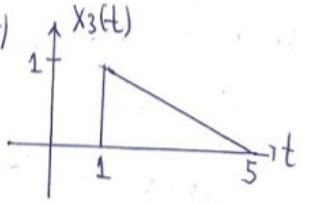
a)



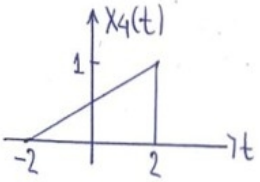
b)



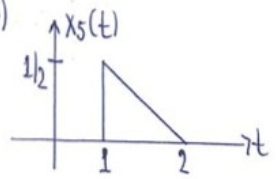
c)



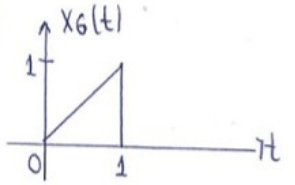
d)



e)

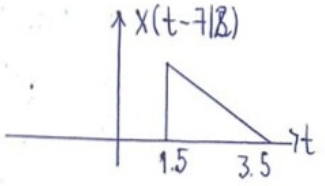


f)



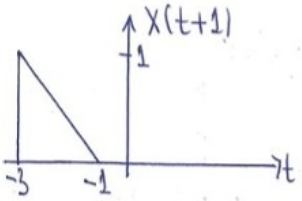
Zgjidhje

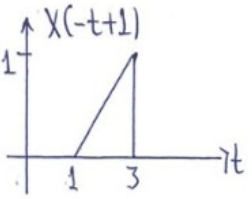
a) Sinjali $x_1(t)$ fitohet duke e zhvendosur sinjalin $x(t)$ për 3.5 njësi djathtas:

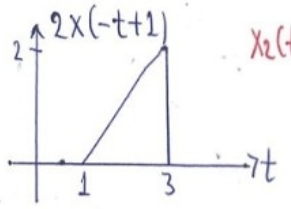


$x_1(t) = x(t-7/2)$

b) Sinjali $x_2(t)$ fitohet duke e zhvendosur sinjalin $x(t)$ për 1 njësi majtas, pastaj duke e pasqyruar dhe në fund duke e amplifikuar në amplitudë për vlerën 2.

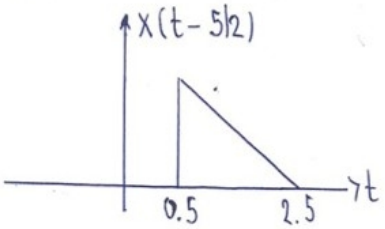


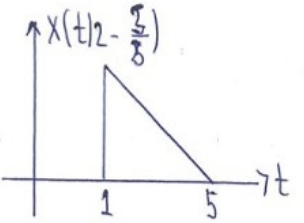




$x_2(t) = 2x(-t+1)$

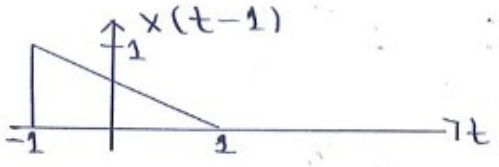
c) Sinjali $x_3(t)$ fitohet duke e zhvendosur sinjalin $x(t)$ për 2.5 njësi djathtas dhe më pas duke e zgjeruar për 2 herë

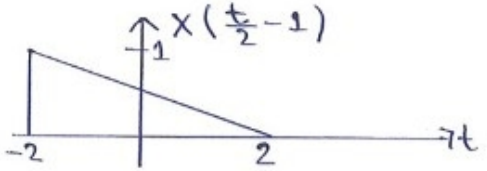


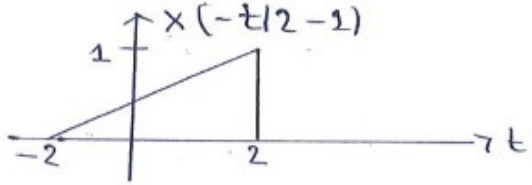


$x_3(t) = x\left(\frac{t}{2} - \frac{5}{2}\right)$

d) Sinjali $x_4(t)$ fitohet duke e zhvendosur sinjalin $x(t)$ për 1 njësi djathtas, pastaj duke e zgjeruar në kohë dhe duke e pasqyruar.

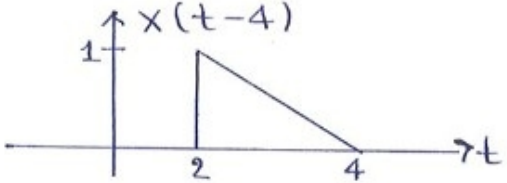


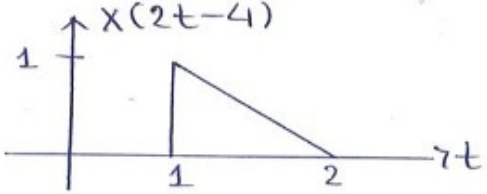


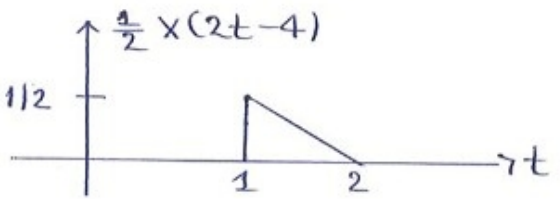


$$x_4(t) = x(-t/2+1)$$

e) Sinjali $x_5(t)$ fitohet duke e zhvendosur sinjalin $x(t)$ për 4 njësi djathtas, duke e ngushtuar për 2 herë dhe në fund duke e dobësuar në amplitudë për vlerën 1/2.

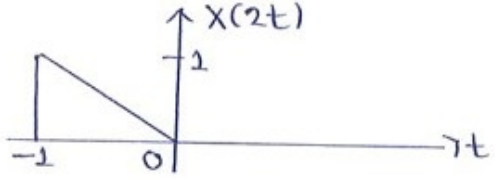


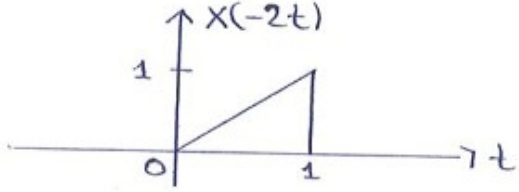




$$x_5(t) = \frac{1}{2}x(2t-4)$$

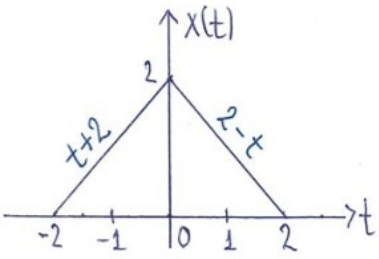
f) Sinjali $x_6(t)$ fitohet duke e ngushtuar sinjalin $x(t)$ për 2 njësi dhe më pas duke e pasqyruar

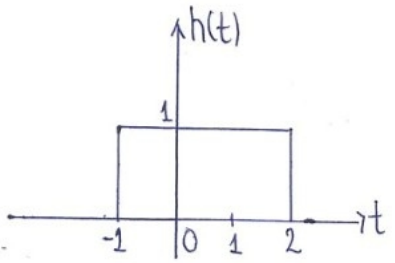




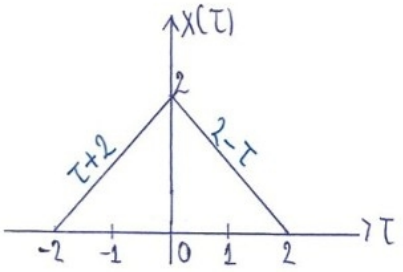
$$x_6(t) = x(-2t)$$

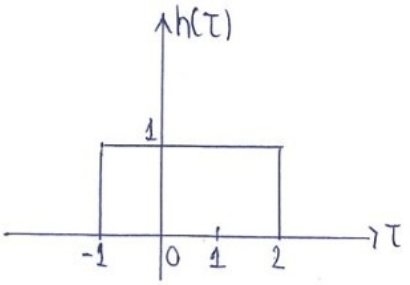
Detyra 2: Përmes thurjes përcakto dhe skico sinjalin në dalje të sistemit nëse hyrja dhe përgjigjja impulsive janë dhënë si më poshtë:



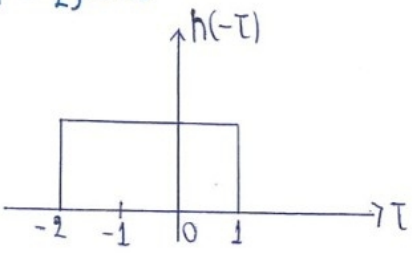


- kalimi nga domeni kohor në domenin τ:

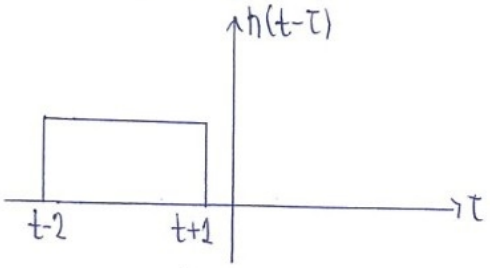




- pasqyrimi:



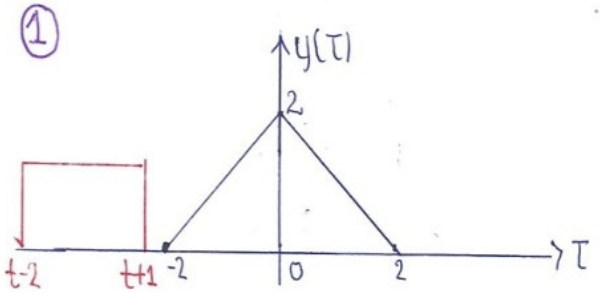
- zhvendosja:



$$y(t)=x(t)*h(t)=\int_{-\infty}^{\infty}x(\tau)h(t-\tau)d\tau$$

Shqyrtimi i rasteve:

①

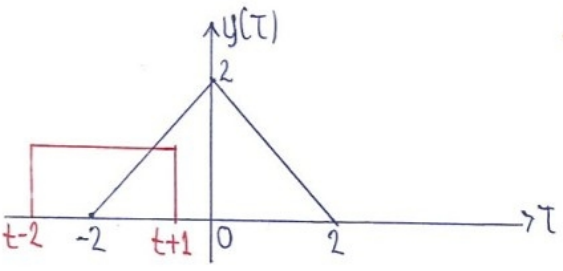


Për: $t+1<-2$

$t<-3$

$y(t)=0$

②

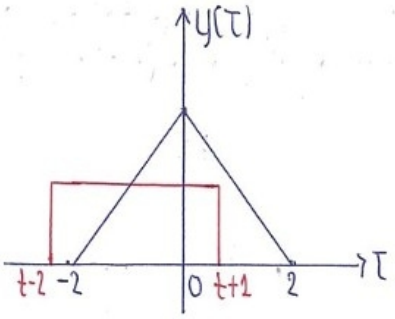


Për $-3<t<-1$; nga figura $t+1<0$, $t<-1$

$$y(t)=\int_{-2}^{t+1}(\tau+2)\cdot 1\,d\tau=\frac{(\tau+2)^2}{2}\Big|_{-2}^{t+1}$$

$$y(t)=\frac{(t+1+2)^2}{2}-0=\frac{(t+3)^2}{2}$$

③



nga figura:

Për $-1<t<0$; $t-2<-2$

$t<0$

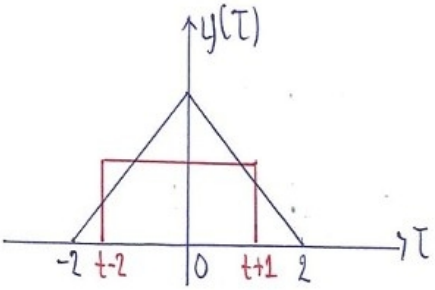
$$y(t)=\int_{-2}^{0}(\tau+2)\cdot 1\,d\tau+\int_{0}^{t+1}(2-\tau)\cdot 1\,d\tau$$

$$y(t)=\frac{(\tau+2)^2}{2}\Big|_{-2}^{0}-\frac{(\tau-2)^2}{2}\Big|_{0}^{t+1}$$

$$y(t)=2-0-\frac{(t-1)^2}{2}+2=4-\frac{1}{2}(t^2-2t+1)$$

$$y(t)=-\frac{t^2}{2}+t+\frac{7}{2}$$

④



nga figura:

Për $0<t<1$; $t+2<2$

$t<1$

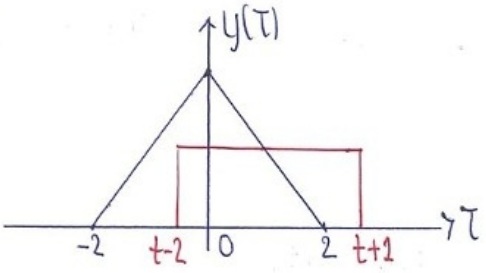
$$y(t)=\int_{t-2}^{0}(\tau+2)\,d\tau+\int_{0}^{t+1}(2-\tau)\,d\tau$$

$$y(t)=\frac{(\tau+2)^2}{2}\Big|_{t-2}^{0}-\frac{(\tau-2)^2}{2}\Big|_{0}^{t+1}$$

$$y(t)=2-\frac{t^2}{2}-\frac{(t-1)^2}{2}+2=4-\frac{t^2}{2}-\frac{t^2}{2}+t-\frac{1}{2}$$

$$y(t)=-t^2+t+\frac{7}{2}$$

⑤



nga figura:

Për $1<t<2$; $t-2<0$

$t<2$

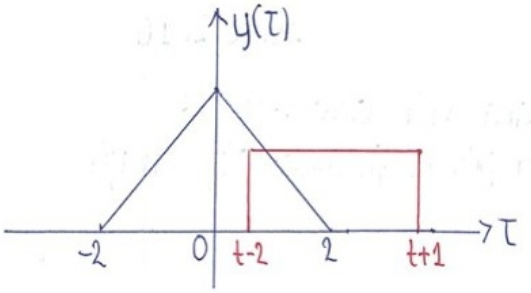
$$y(t)=\int_{t-2}^{0}(\tau+2)\,d\tau+\int_{0}^{2}(2-\tau)\,d\tau$$

$$y(t)=\frac{(\tau+2)^2}{2}\Big|_{t-2}^{0}-\frac{(\tau-2)^2}{2}\Big|_{0}^{2}$$

$$y(t)=2-\frac{t^2}{2}-0+2$$

$$y(t)=4-\frac{t^2}{2}$$

⑥



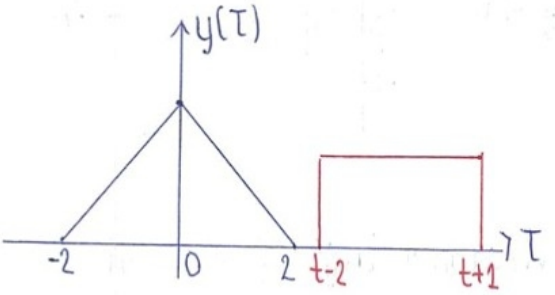
Për $2<t<4$; $t-2<2$, $t<4$

$$y(t)=\int_{t-2}^{2}(2-\tau)d\tau=-\frac{(\tau-2)^2}{2}\Big|_{t-2}^{2}$$

$$y(t)=\frac{(t-4)^2}{2}=\frac{1}{2}(t^2-8t+16)$$

$$y(t)=\frac{t^2}{2}-4t+8$$

⑦



Për $t>4$

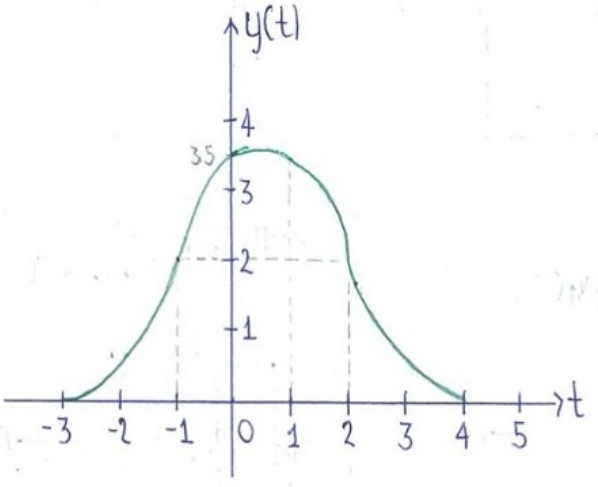
$y(t)=0$

Paraqitja grafike:

$$y(t)=\begin{cases}0 & t<-3\\ (t+3)^2/2 & -3<t<-1\\ -t^2/2+t+7/2 & -1<t<0\\ -t^2+t+7/2 & 0<t<1\\ 4-t^2/2 & 1<t<2\\ t^2/2-4t+8 & 2<t<4\\ 0 & t>4\end{cases}$$

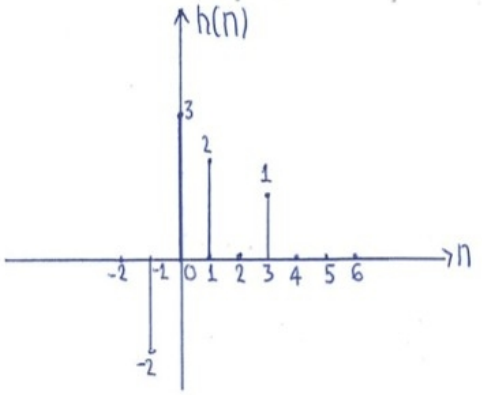
Sqarim:

ax^2+bx+c paraqet ekuacionin e parabolës.



Detyra 19:

Një sistem diskret, linear dhe invariant në zhvendosje, ka përgjigje impulsive si në figurë:



Të përcaktohet sinjali në dalje y(n) për këto dy raste të x(n):

a) $x(n) = 3\delta(n) - 2\delta(n-1)$
b) $x(n) = 3u(n+1) - u(n-3)$

Zgjidhje

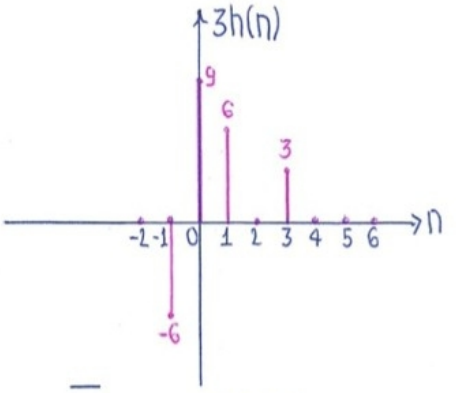
Metoda e parë:

a) Thurja: $y(n) = x(n) * h(n)$

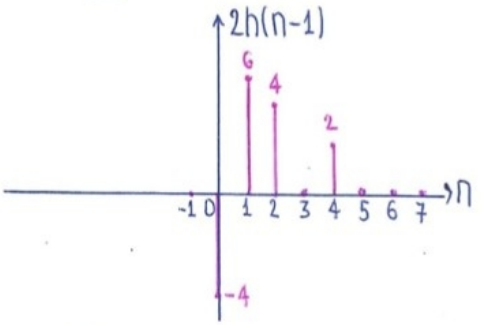
$y(n) = [3\delta(n) - 2\delta(n-1)] * h(n)$

$y(n) = 3\delta(n) * h(n) - 2\delta(n-1) * h(n)$

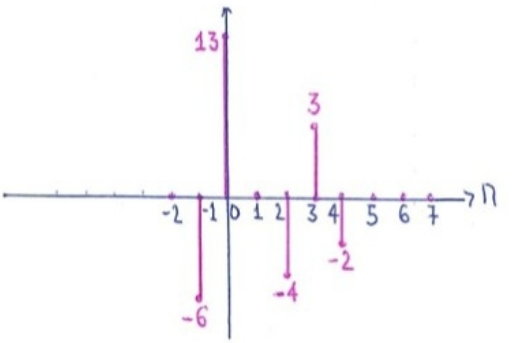
$y(n) = 3h(n) - 2h(n-1)$



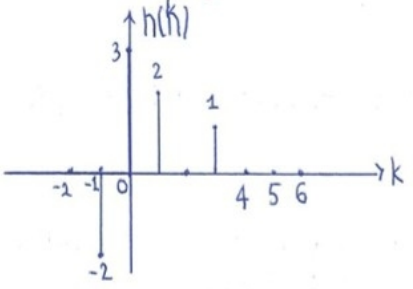
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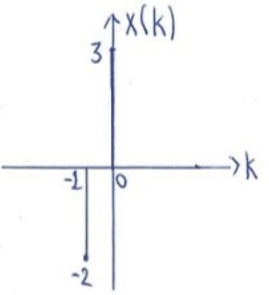


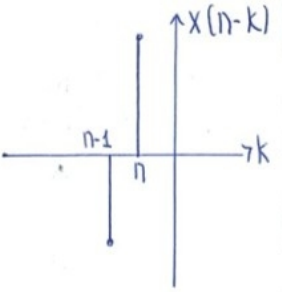
=



Metoda e dytë: (mostër për mostër)







① Për $n<-1$ $\quad y(n)=0$

② Për $n=-1$

$$y(n)=\sum_{k=-1}^{1} h(k)x(-1-k)$$

$$y(-1)=h(-1)x(0)=-2\cdot 3$$

$$y(-1)=-6$$

③ Për $n=0$

$$y(n)=\sum_{k=-1}^{0} h(k)x(-k)$$

$$y(0)=h(-1)x(1)+h(0)x(0)$$

$$y(0)=-2\cdot(-2)+3\cdot 3=4+9$$

$$y(0)=13$$

④ Për $n=1$

$$y(n)=\sum_{k=0}^{1} h(k)x(1-k)$$

$$y(1)=h(0)x(1)+h(1)x(0)$$

$$y(1)=3\cdot(-2)+2\cdot 3=-6+6$$

$$y(1)=0$$

⑤ Për $n=2$

$$y(n)=\sum_{k=1}^{2} h(k)x(2-k)$$

$$y(2)=h(1)x(1)+h(2)x(0)$$

$$y(2)=2\cdot(-2)+0\cdot 3$$

$$y(2)=-4$$

⑥ Për $n=3$

$$y(n)=\sum_{k=2}^{3} h(k)x(3-k)$$

$$y(3)=h(2)x(1)+h(3)x(0)$$

$$y(3)=0\cdot(-2)+1\cdot 3$$

$$y(3)=3$$

⑦ Për $n=4$

$$y(n)=\sum_{k=3}^{4} h(k)x(4-k)$$

$$y(4)=h(3)x(1)+h(4)x(0)$$

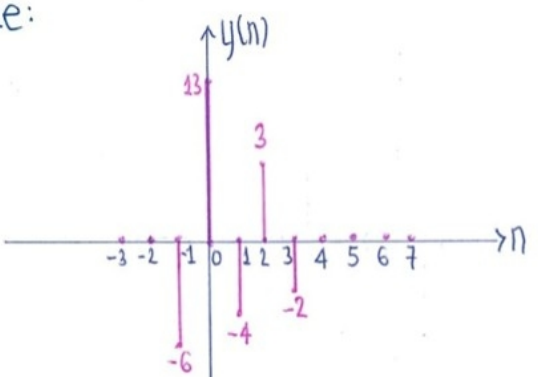
$$y(4)=1\cdot(-2)+0\cdot 3$$

$$y(4)=-2$$

⑧ Për $n>4$

$$y(n)=0$$

Paraqitja grafike:



b) $x(n) = 3u(n+1) - u(n-3)$

$y(n) = T\{x(n)\}$

$y(n) = T\{3u(n+1) - u(n-3)\}$

$y(n) = 3T\{u(n+1)\} - T\{u(n-3)\}$

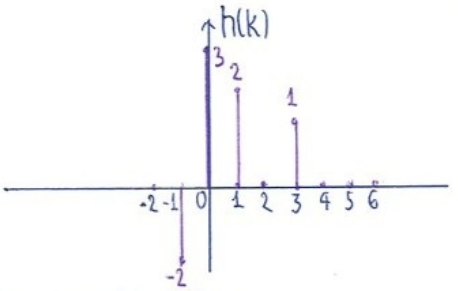
$y(n) = 3S(n+1) - S(n-3)$

$S(n)$ e gjejmë duke thurur përgjigjen impulsive me sinjali shkallë njësi.

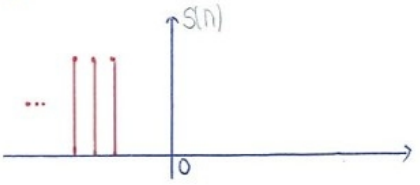
Thurja: $S(n) = h(n) * u(n) = \sum_{k=-\infty}^{\infty} h(k)u(n-k) = \sum_{k=0}^{n} h(k)$

$n-k \geq 0$

$n \geq k$

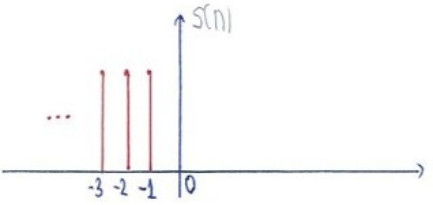


Shqyrtimi i rasteve:

① 

Për $n < -1$;

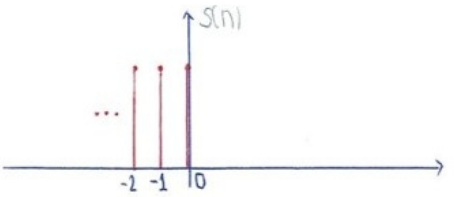
$S(n) = 0$

② 

Për $n = -1$;

$S(-1) = \sum_{k=-1}^{1} h(k) = h(-1)$

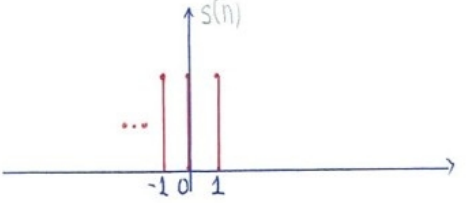
$S(1) = -2$

③ 

Për $n = 0$;

$S(0) = \sum_{k=-1}^{0} h(k) = h(-1) + h(0) = -2 + 3$

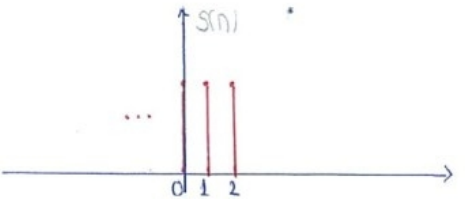
$S(0) = 1$

④ 

Për $n = 1$;

$S(1) = \sum_{k=-1}^{1} h(k) = h(-1) + h(0) + h(1) = -2 + 3 + 2$

$S(1) = 3$

⑤ 

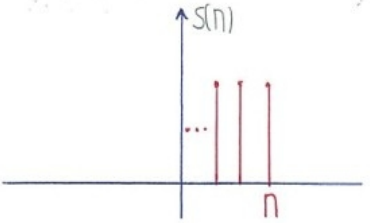
Për $n = 2$

$S(2) = \sum_{k=-1}^{2} h(k) = h(-1) + h(0) + h(1) + h(2)$

$S(2) = -2 + 3 + 2 + 0$

$S(2) = 3$

⑥



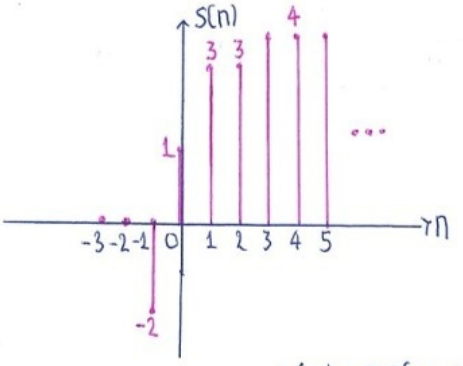
Për $n \geq 3$

$$s(n) = \sum_{k=-1}^{n} h(k) = h(-1)+h(0)+h(1)+h(2)+h(3)$$

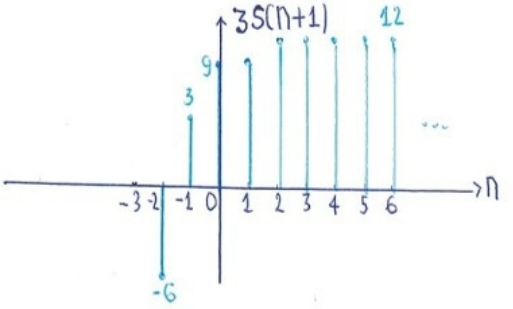
$$s(n) = -2+3+2+0+1 = 4$$

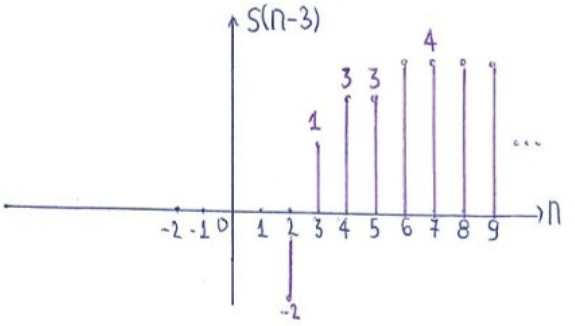
$$s(n) = 4$$

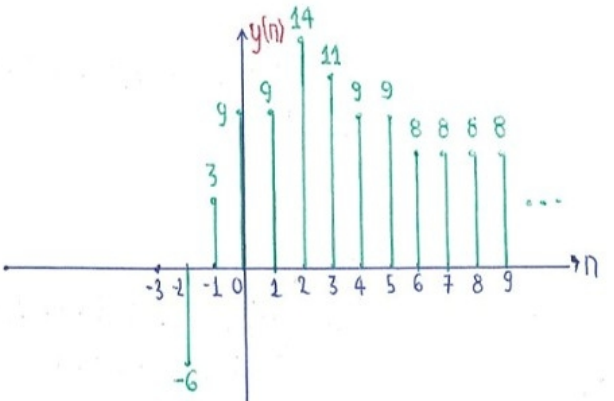
Paraqitja grafike:



$\longrightarrow$ $y(n) = 3s(n+1) - s(n-3)$







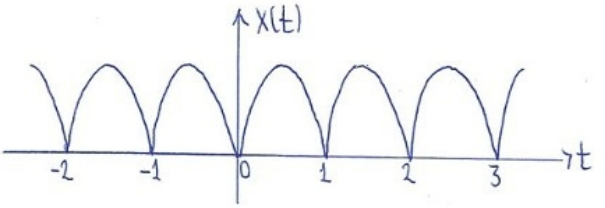
Shembulli 5:

Është dhënë sinjali $x(t) = |\sin \pi t|$

a) Të skicohet sinjali $x(t)$
b) Të shënohet forma eksponenciale e serisë furie
c) Të shënohet forma trigonometrike e serisë furie

Zgjidhje

a) Paraqitja grafike e sinjalit të dhënë:



b) Perioda e sinjalit të dhënë: $T = 1$; $\omega = \frac{2\pi}{T} = 2\pi$

Forma eksponenciale e serisë furie: $x(t) = \sum_{k=-\infty}^{\infty} C_k e^{jk\omega_0 t}$

$$C_k = \frac{1}{T}\int_T x(t) e^{-jk\omega_0 t} dt$$

Në rastin tonë kemi:

$$C_k = \frac{1}{1}\int_0^1 x(t) e^{-jk2\pi t} dt = \int_0^1 \sin \pi t \, e^{-jk2\pi t} dt$$

$$C_k = \int_0^1 \frac{e^{j\pi t} - e^{-j\pi t}}{2j} \cdot e^{-jk2\pi t} dt$$

$$C_k = \frac{1}{2j}\int_0^1 \left[e^{-j\pi(2k-1)t} - e^{-j\pi(2k+1)t}\right] dt$$

$$C_k = \frac{1}{2j}\left[\frac{1}{-j\pi(2k-1)} e^{-j\pi(2k-1)t}\Big|_0^1 - \frac{1}{-j\pi(2k+1)} e^{-j\pi(2k+1)t}\Big|_0^1\right]$$

$$C_k = -\frac{1}{2\pi}\left[\frac{1 - e^{-j\pi(2k-1)}}{2k-1} - \frac{1 - e^{-j\pi(2k+1)}}{2k+1}\right]$$

$e^{-j\pi 2k} = (e^{-j\pi})^{2k} = (-1)^{2k} = +1$

$e^{\mp j\pi} = -1$

$$C_k = -\frac{1}{\pi}\left[\frac{2k+1-2k+1}{4k^2-1}\right]$$

$$C_k = \frac{-2}{\pi(4k^2-1)} = \frac{2}{\pi(1-4k^2)}$$

Për $k=0$:

$$C_0 = \frac{2}{\pi(1-4\cdot 0^2)} = \frac{2}{\pi}$$

$$a_k = 2\,\mathrm{Re}\{C_k\} = \frac{4}{\pi(1-4k^2)}$$

$$b_k = -2\,\mathrm{Im}\{C_k\} = 0$$

$$\frac{a_0}{2} = C_0 = \frac{2}{\pi}$$

$$x(t) = \frac{2}{\pi} + \sum_{k=1}^{\infty} \frac{4}{\pi(1-4k^2)} \cos(2\pi k)t$$

Nëse do të ishte $\frac{4}{\pi(k^2-1)}$ atëherë duhet gjetur C_1 dhe C_{-1}.

$$C_1 = \frac{1}{1}\int_0^1 x(t)\, e^{-j\cdot 1\cdot 2\pi t}.$$

ku $C_{-1} = C_1^*$